

Translated Version of the Document (English)

DATABASES – SEMESTER PROJECT (2025–2026)

National Technical University of Athens (NTUA)
School of Electrical and Computer Engineering
Knowledge and Data Systems Laboratory

I. Assignment

The “Ygeiopolis” General Hospital has assigned you the task of designing and implementing a system for storing and managing the information required for its operation.

The hospital staff includes doctors, nurses, and administrative personnel. Every staff member has a social security number (AMKA), first name, last name, age, email, phone number, hiring date, and staff type.

Doctors additionally have:

- Medical association license number
- Specialty (e.g., Cardiology, Surgery)
- Rank (Resident, Consultant B', Consultant A', Director)

Each doctor may have a supervisor, who is also a doctor in the hospital. Residents must have a supervisor, while Directors do not. Cyclic supervision chains are prohibited.

Doctors may belong to one or more departments. Nurses have ranks (Assistant Nurse, Nurse, Head Nurse) and belong to one department. Administrative staff have duties/roles (e.g., Secretary, Accountant), office location, and belong to one department.

The hospital has departments such as Cardiology, Surgery, ICU, and Emergency. Each department has a name, description, number of beds, floor/building, and a department director (doctor). Each department contains beds with a unique number, type (e.g., ICU, single-bed, multi-bed), and status (available, occupied, under maintenance).

Patients have:

- AMKA
- First and last name
- Father's name
- Age
- Gender
- Weight and height
- Address
- Phone and email
- Profession
- Nationality
- Emergency contact information
- Insurance provider

Each patient may have multiple hospitalizations in their history.

All hospital departments operate on duty 24 hours a day. Their operation is organized into three 8-hour shifts:

- Morning (07:00–15:00)
- Afternoon (15:00–23:00)
- Night (23:00–07:00)

Each department shift must be staffed by:

- At least 3 doctors
- At least 6 nurses
- At least 2 administrative employees

If a resident participates in a shift, at least one Consultant A' or Director must also be present.

Maximum monthly shifts:

- Doctors: 15
- Nurses: 20
- Administrative staff: 25

At least 8 hours of rest must separate consecutive shifts for the same employee. No employee may participate in more than 3 consecutive night shifts.

Emergency patients first pass through triage, where a nurse records symptoms and assigns an urgency level:

- 1 = immediate
- 2 = urgent
- 3 = pressing
- 4 = less urgent
- 5 = non-urgent

Patients are served based on urgency and FIFO ordering within the same urgency level.

Patients may either:

- Receive instructions and leave
- Be admitted for hospitalization

Hospitalization records include:

- Patient
- Bed
- Department
- Admission and discharge dates
- Admission and discharge diagnosis (ICD-10 code and description)
- Total hospitalization cost

Costing follows the DRG/KEN system. Each hospitalization corresponds to a KEN code defining the base cost and expected average duration of stay. Additional daily charges apply if the stay exceeds the expected duration.

Hospitalizations may include:

- Laboratory tests
- Medical procedures/interventions
- Medication prescriptions

Each laboratory test has:

- Code
- Type
- Date
- Result
- Cost
- Ordering physician

Each medical procedure has:

- Code

- Name
- Category
- Duration
- Cost
- Required room

A procedure is performed by one lead surgeon and optionally assistants. Two procedures cannot occur simultaneously in the same room, and the same doctor cannot participate in multiple simultaneous procedures.

Medication is prescribed only by doctors. Prescriptions include dosage, frequency, and start/end dates. The combination of doctor, patient, medication, and start date must be unique.

The hospital follows the EMA Article 57 database containing all approved medicines in the European Economic Area.

Patients may have allergies to active substances. Medication containing allergic substances cannot be prescribed.

Patients with completed hospitalization may evaluate:

(a) Hospitalization:

- Nursing care quality
- Cleanliness
- Food
- Overall experience

(b) Doctors:

- Quality of medical care

Patients may evaluate any doctor who prescribed medication during their hospitalization.

All entities should also support associated images and text descriptions for the future hospital website.

II. Implementation Guidelines

You must document assumptions in README.md.

The database must contain enough data to execute all required queries successfully.

Indicative minimum data:

- 80 doctors
- 200 patients
- 500 hospitalizations
- 15 departments
- 300 prescriptions
- 10 operating rooms
- 150 medical procedures
- 200 laboratory tests

Reference datasets (ICD-10, medicines, KEN, procedures) must be imported from official sources without inventing codes or names.

III. Deliverables

A. Database Design

1. Design the ER diagram.
2. Design the relational schema and implement the database.
3. Define integrity constraints, keys, referential integrity, and indexes.

B. SQL Queries

Implement 15 SQL queries including:

- Hospital revenues
- Doctor activity
- Patient hospitalization history
- Ratings analysis
- Shift scheduling
- Allergy statistics
- Recursive supervision hierarchy
- ICD-10 statistics
- Triage analysis

Queries 4 and 6 must additionally include:

- EXPLAIN / EXPLAIN ANALYZE
- Alternative version with FORCE INDEX or hints
- Comparison of estimated cost and execution time
- Short conclusions

IV. Administrative Information

- Teams up to 3 students
- Suggested DBMS: MariaDB/MySQL or PostgreSQL
- ORM technologies are prohibited
- SQL must be elegant and efficient
- AI/LLM tools may be used only if explicitly documented

V. Submission

Submission deadline: 17/5/2026

Required ZIP structure:

- README.md
- diagrams/
- sql/
- docs/report.pdf
- optional code/

VI. Demonstration

A demonstration before exams is mandatory.
Students may be asked additional ad-hoc SQL questions.

VII. Grading

- 90 points: requirements and functionality
- 10 points: creativity and presentation
- Additional 10 points for web UI implementation

VIII. Deadlines

12/3/2026 – Assignment release
12/4/2026 – Team registration
17/5/2026 – Submission
18–22/5/2026 – Demonstration